PRODUCT BRIEF

Cross-currency group expense settlement on Monad: MeshTab

Close the tab. Keep the friends.

October 2026 | Version 1.0

Document Information

Product Name	MeshTab
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Primary Stakeholders	Monad Metropolis judges and sponsor-bounty reviewers
Event / Track	Monad Metropolis Hackathon, Consumer Products & Payments
Submission Date	October 13, 2026
Document Status	Final
Stage	MVP prototype build (Monad Testnet)
Live App	https://mesh-tab.vercel.app
Companion Documents	PRD v1.0 Technical Brief v1.0, README
Last Updated	October 2026

1. SUMMARY

MeshTab is a group-expense app for friends who spend in more than one currency. Members log costs in one place, see exactly who owes whom, and settle with one tap and no gas fee. It closes the gap between "tracking a debt" and "paying it," so a trip ends with one short payment list instead of several lists in several currencies. This release is an unaudited prototype on Monad Testnet, settling in the issuer's official testnet AUSD, with no real money. Judges can test it themselves, with their own numbers, in about five minutes.

2. PROBLEM

Who

Friend groups who share costs across currencies and borders: trip organizers, remote friends, expats and digital nomads. Domestic groups such as roommates and event crews are a secondary segment.

What

Shared spending across borders is fragmented and strains friendships:

- Debts are tracked in one app and paid in another, so the app records but never settles.
- On multi-currency tabs, common expense trackers net debts within each currency separately, leaving a payment list per currency.
- Organizers front the costs and chase repayment. A US consumer survey found 76% of Gen Z who fronted group costs were not fully repaid (PRD Appendix B).
- Cross-border repayment can add foreign-transaction fees and FX markups (hypothesis, not quantified).
- Crypto payment tools expect wallets, seed phrases and gas tokens, so non-crypto friends will not use them (hypothesis, tested in validation).

Why now

- Passkeys and smart accounts let a wallet exist without a seed phrase, and paymasters remove the gas step.
- Monad's published numbers (400 ms blocks, 800 ms finality) make on-chain settlement feel close to instant.
- Stablecoin rails are live on Monad mainnet. This prototype runs on testnet so evaluators can try it with no real money.
- Passkey sign-in works on the phones people already carry.

3. TARGET USER and PRIMARY PERSONA

Segments

- **Primary:** cross-border and travel groups. Personas: Aisha, Kemi, Ben, Danny.
- **Secondary:** nomads and coordinators. Personas: Sarah, Liam, Tunde.
- **Adjacent (acknowledged, not designed for):** Chloe, Alex, Elena. Their needs are only partly met, and the PRD states the gaps.

Primary persona: "Aisha, the Trip Organizer"

Name	Aisha Bello, 27
Role	Senior Product Designer, Lagos
Tools today	Group chat, a notes or spreadsheet tab, an expense tracker, and a separate payment app
Job-to-be-done	"When the trip ends, help me get everyone paid back correctly without chasing anyone."
Success looks like	A payment list she trusts in under a minute, and friends who pay in one tap

4. SOLUTION

A mobile-first web app built around three loops:

Log: any member records an expense in one of 9 currencies by hand, receipt photo, or voice (where the browser supports it). The dollar value is fixed at entry. Logging works offline.

Net: the app combines everyone's balances across currencies into at most (non-zero members minus 1) payments, and shows the math: each member's net position, a before/after graph, and a $0.00 reconciliation check.

Settle: the debtor taps Confirm Settlement and confirms on their device (a biometric check, or a slide or typed amount when no passkey is enrolled). One user operation batches the approval and every payment, a Pimlico paymaster pays testnet gas, and an explorer receipt confirms it in about 1.3 seconds. Funds come from the issuer's official testnet faucet in one tap.

Illustration (not a promise): four friends, six expenses in EUR, GBP and USD. Netting by currency would leave 9 payments. MeshTab produces 3, in one currency. Counts change with the number of people, currencies and who paid (PRD Appendix A).

Design principles

- **Simplicity over crypto:** no wallets, seed phrases, gas or chain jargon in primary flows.
- **Trust by design:** every change is attributed, the math is visible, balances reconcile to $0.00, and each payment is verifiable on the explorer.
- **Honest by default:** labeled testnet, labeled testnet token, labeled indicative rates. No claims of an audit,

custody or a trustless ledger.

- **Works on the phones people already have:** web app, passkeys, no extension or hardware.
- **Free to try:** a sample tab with no sign-in, plus a one-tap funds claim, so judges can verify claims with their own numbers.

5. MVP SCOPE

In scope (MVP)	Out of scope (this release)
• Passkey, Google or email sign-in with automatic smart wallet • Create and join tabs by link or code. Up to 15 members per tab, 20 maximum • Expense logging in 9 currencies (6 live rates, 3 indicative) with four split types • Receipt scan into an editable draft • Cross-currency netting with visible math • One-tap sponsored settlement in the issuer's official testnet AUSD • One-tap test funds from the issuer's official faucet • Offline expense drafting and netting • Sample and testnet modes • Judge self-test path	• Mainnet and the production stablecoin • Fiat on- and off-ramps • Virtual card and yield vault (prototype UI only) • Group-atomic settlement in one transaction • Payment enforcement and escrow • Offline settlement • Push, email and SMS reminders • Security audit • Native mobile apps

6. SUCCESS METRICS

North Star

Payments eliminated per tab versus a per-currency baseline. It captures the core value (fewer payments) and is testable on any tab. The guarantee is a property: payments never exceed non-zero members minus 1.

Input metrics

- **Activation:** median time to first split under 30 seconds.
- **Engagement:** 90% or more of testers who open Settle Up complete it.
- **Accuracy:** balances sum to exactly $0.00 and payments settle every position in 100% of randomized test tabs.

Output metrics

- **Speed:** measured median tap-to-confirmed time is 1.28 s (slowest 1.54 s) across 10 real testnet settlements on 2 devices.
- **Comprehension:** 4 of 5 testers correctly explain who pays whom.
- **Self-test:** 4 of 5 first-time testers complete the verify-it-yourself path in five minutes.

Guardrails

- User-paid gas stays at $0.
- No offline operation is silently dropped.
- No claim appears in any document or screen without evidence.

7. ASSUMPTIONS and RISKS

Critical Testing Assumptions (Hackathon Phase)

- **Testnet scope:** evaluators use free testnet tokens with no value. Nothing in the product or documents implies real money.
- **Illustrative figures:** demo numbers are examples. Outcomes depend on members, currencies and spend. The product guarantees properties, not fixed counts.

Key assumptions

Assumption	How we validate
Non-crypto friends will sign in with a passkey and pay without wallet knowledge	5 moderated sessions: time to first split, Settle Up completion, and a post-session trust question
Cross-currency netting meaningfully reduces payments on multi-currency tabs	Simulation over randomized tabs, plus the benchmark compared against a real per-currency result
One-tap sponsored settlement feels fast enough	10 timed settlements on 2 devices
Receipt scan saves time	Timed scans of 10 receipts
Judges can reproduce our claims themselves	Self-test path run by first-time users

Top risks and mitigations

- **Claims outrun evidence:** every claim maps to an item in the PRD's Verification Register. Figures are labeled illustrative.
- **Issuer faucet unavailable or rate-limited during judging:** test it daily, keep a pre-claimed reserve of testnet AUSD, and show a clear message if a claim fails.
- **Hybrid architecture questioned:** state plainly that payments are on-chain and netting and history are off-chain.
- **Provider outages during judging (network, paymaster, sign-in):** pre-fund the paymaster, set a spend-ceiling alert, show a status banner, keep Google and email fallbacks, and keep a recorded demo.
- **Competitive pressure from Web2 expense trackers and Web3 payment apps:** win on cross-currency netting plus in-app settlement with zero crypto friction, not feature breadth. On single-currency tabs the payment count does not differ, so we claim only the settlement edge there.
- **FX rate trust:** three currencies use hardcoded estimates. They are labeled indicative in the UI.
- **Real-money compliance:** out of scope now. Legal review is required before any mainnet launch.

8. ROADMAP SIGNAL (POST-MVP)

- **Mainnet and security review (Q4 2026):** a pre-mainnet security review (an independent audit, plus a security scan if we are selected for the ack3 sponsor prize), then mainnet with the production stablecoin.
- **Automated settlement:** opt-in session keys with spend caps, so a group can pre-approve settlement.
- **Fiat funding and cash-out (Q1 2027):** through Privy (Stripe on-ramp), subject to provider coverage of networks, assets and currencies. Local-currency cash-out starts with Nigeria.
- **Virtual debit card:** spend settled stablecoin balances at Visa merchants through Privy's card program, subject to program access and supported assets.
- **Group vault:** idle group funds earning yield, once a custody design exists.
- **Reminders and enforcement:** push, email and SMS reminders (an in-app nudge exists), and optional escrow for trip deposits.
- **Native mobile apps** with background sync.

9. OPEN QUESTIONS and STRATEGIC DECISIONS

Resolved Decisions

- **Network:** Monad Testnet only for this release, so evaluators can test safely.
- **Token:** settlement uses the issuer's official testnet AUSD (6 decimals), claimed from the issuer's own faucet (10,000 per claim, 100,000 wallet ceiling). The full flow is proven on testnet with public transactions. The ledger fixes its token address at deployment, so mainnet means a new deployment.
- **Settlement design:** one payer-to-recipient transfer per call. No member can move another member's funds, so group-atomic settlement is out of scope.
- **Capacity:** 15 members per tab supported, 20 as the hard ceiling.
- **Gas sponsorship:** whitelisted contracts only, a monthly spend ceiling with an alert, and no per-user cap on testnet.
- **Signing:** devices with an enrolled passkey confirm each settlement with a biometric check. Devices without one confirm by slide or typed amount, and the session signs. Acceptable on testnet. Real-money use will require re-authentication or limits.
- **Positioning:** cross-border groups lead. Generic splitting is secondary.
- **Offline scope:** logging and netting work offline. Settlement does not.
- **Netting claim:** greedy net-position matching with at most (non-zero members minus 1) payments. We do not claim the theoretical minimum.
- **Naming and sponsors:** other products are referred to by category. We name only what we use. All claims rest on public documentation.

Current Open Questions

- Does the 1.3-second timing hold for settlements with several creditors? Only a single payment was timed.

Post-Launch Focus

- Which partners cover our first markets, including NGN, for on- and off-ramps?
- Is the business model a small settlement fee, a premium tier, or partner revenue?
- Escrow versus reputation: how much enforcement do groups actually want?

10. ECOSYSTEM ALIGNMENT

Partner	Role in MeshTab	Status
Monad	Settlement network and ledger contract	Live on testnet
Agora	Official testnet AUSD and faucet used for all settlement. Production token is roadmap	Live on testnet
Privy	Passkeys and ERC-4337 smart wallets. Planned: funding and card	Live (auth). Roadmap (funding, card)
Pimlico	Paymaster that sponsors testnet gas	Live on testnet
Pyth Network	Live FX for 6 currencies (3 more are indicative)	Live
Google Gemini	Receipt scan into an editable draft	Live
Upstash Redis	Off-chain tab state and sync	Live
ack3	Security scan for winning teams	Roadmap, if selected